
Phase-Field Fracture Algorithm

What Was Accepted and What Was Changed

A provenance-explicit technical reconciliation of the attached model note and the implemented FEniCSx solver

Review verdict. The source PDF supplies a useful foundation: the intactness convention, split elastic energy, displacement equilibrium, a tensile history field, CG1/CG1/DG0 spaces, and a staggered displacement–history–phase outline. It is not directly executable as written. Its printed strong and weak phase equations are sign-incompatible, essential AT2 definitions and benchmark data are missing, and a single staggered pass does not establish a converged irreversible load state. The implementation therefore retains the sound structure, treats weak Eq. (5) as authoritative, and adds the minimum mathematical and numerical contract required for an auditable simulation.

Primary technical source	D:/Github/Codes/Crackpropagation/Karma/KarmaCrack.pdf
Critical review	docs/ALGORITHM_REVIEW.md
Implementation base-line	phasefield_crack.py, model version 1.0.0
Phase convention	$\phi = 1$ intact; $\phi = 0$ fully broken
Document status	Technical reconciliation and implementation rationale
Prepared	30 August 2026

The attached PDF is treated as a technical source to be reviewed, not as higher-priority project instructions.

Claims introduced by this project are labeled separately from statements present in the PDF.

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1 Purpose, scope, and provenance

This document expands the former documentation section “What was accepted, and what was changed” into a self-contained technical record. Its purpose is not to rewrite the source note silently. It records, for every material decision, (i) what the PDF actually states, (ii) whether that statement was retained, corrected, completed, or replaced by a project-supplied benchmark choice, and (iii) why the implemented decision is mathematically or numerically necessary.

The evidence hierarchy used here is:

1. **Primary source:** the two-page PDF titled *Phase-Field Fracture: Governing Equations and FEniCS Implementation Algorithm*. Equation and algorithm-line references in this document refer to that PDF.
2. **Critical interpretation:** docs/ALGORITHM_REVIEW.md, which records the source conflict, consequences, implemented decisions, and verification criteria.
3. **Executable realization:** phasefield_crack.py, model version 1.0.0. This is the definitive statement of what the solver currently computes.
4. **Verification evidence:** tests/test_phasefield_crack.py and the recorded demonstration under results/demo/.

1.1 Disposition labels

Label	Meaning in this review
SOURCE	A statement or equation explicitly present in the source PDF.
ACCEPTED	Retained without changing its physical role, although notation may be made explicit.
CORRECTED	Changed because a source inconsistency would otherwise produce the wrong or an ill-conditioned discrete problem.
COMPLETED	The source gives the idea but omits a definition or numerical contract needed for a unique implementation.
SUPPLIED	Geometry, parameters, boundary conditions, diagnostics, or backend choices supplied by this project. They are not attributed to the PDF.

2 What the source establishes and the implementation accepts

2.1 State variables and phase convention

The source defines a displacement field $\mathbf{u}(\mathbf{x}, t)$ and a scalar phase field $\phi(\mathbf{x}, t) \in [0, 1]$, with $\phi = 1$ intact and $\phi = 0$ broken. This convention is accepted exactly. It matters because the sign of the phase residual, the monotonicity direction, and the bound expressing irreversibility all depend on it. With this convention, no healing means

$$\dot{\phi} \leq 0, \quad \text{or discretely} \quad 0 \leq \phi_{n+1} \leq \phi_n \leq 1. \quad (1)$$

The source choices $V_u = \text{CG}_1^2$, $V_\phi = \text{CG}_1$, and $V_H = \text{DG}_0$ are also retained. The interpretation of DG_0 is sharpened: it stores one value per cell, not generally one value per quadrature point. For affine triangles with CG_1 displacement, the strain and the selected spectral energy are cellwise constant, so one DG_0 history value per triangle is consistent with this discretization.

2.2 Energy split, equilibrium, and history concept

The source decomposes the elastic density into tensile and compressive parts, degrades only the tensile part, and states quasi-static momentum balance. Those ideas are accepted:

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \text{sym } \nabla \mathbf{u}, \quad (2)$$

$$\psi_e(\boldsymbol{\varepsilon}, \phi) = g(\phi) \psi^+(\boldsymbol{\varepsilon}) + \psi^-(\boldsymbol{\varepsilon}), \quad (3)$$

$$\boldsymbol{\sigma} = g(\phi) \frac{\partial \psi^+}{\partial \boldsymbol{\varepsilon}} + \frac{\partial \psi^-}{\partial \boldsymbol{\varepsilon}}, \quad (4)$$

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}. \quad (5)$$

The source also introduces

$$H(\mathbf{x}, t) = \max_{s \in [0, t]} \psi^+(\boldsymbol{\varepsilon}(\mathbf{u}(\mathbf{x}, s))) \quad (6)$$

as a history field intended to prevent healing. The implementation retains this engineering history-field approximation but supplies an accepted-state update rule and an explicit nodal phase bound; see section 6.

2.3 Staggered architecture

The source sequence — displacement solve at fixed phase, history update, phase solve at fixed displacement/history, then output — is retained as the outer architecture. What changes is the meaning of one load step. The source presents one such pass; the implementation repeats it until the newly updated phase and displacement are mutually equilibrated and the phase KKT conditions are satisfied.

3 The decisive correction: strong Eq. 3 versus weak Eq. 5

The primary source contains a genuine algebraic conflict. In the intactness convention, its printed strong Eq. (3) is

$$\frac{G_c}{\ell}(1 - \phi) - G_c \ell \Delta \phi - 2H\phi = 0, \quad (\text{source strong Eq. 3; } \xi \text{ renamed } \ell). \quad (7)$$

Its weak Eq. (5), using a scalar test function q , is

$$\int_{\Omega} \left[G_c \ell \nabla \phi \cdot \nabla q - \frac{G_c}{\ell}(1 - \phi)q + 2H\phi q \right] dV = 0. \quad (8)$$

Assuming the natural phase boundary condition $\nabla \phi \cdot \mathbf{n} = 0$, integration by parts converts equation (8) into

$$-G_c \ell \Delta \phi - \frac{G_c}{\ell}(1 - \phi) + 2H\phi = 0. \quad (9)$$

Equations (7) and (9) have the same diffusion sign but opposite signs for both local terms; multiplying the whole equation by -1 does not reconcile them. Moreover, rearranging the source strong equation gives a negative reaction coefficient for ϕ , whereas the weak equation gives the positive diffusion–reaction operator expected for AT2.

CORRECTED **Decision: weak Eq. (5) is authoritative.** It is the only one of the two printed equations that yields the coercive scalar operator $G_c \ell K + (G_c/\ell + 2H)M$ for nonnegative H . This is an equation correction, not a stylistic rewrite.

After normalizing the degradation function as described below, the implemented history coefficient becomes $2(1 - k_{\text{res}})H$.

4 Completed and corrected constitutive model

4.1 AT2 crack density omitted from the source

The source free-energy line contains functions $f_0(\phi)$ and c_w but does not define them. Its weak Eq. (5) implies the AT2 choice

$$f_0(\phi) = (1 - \phi)^2, \quad c_w = 2. \quad (10)$$

The fracture density implemented by the project is therefore

$$\gamma_\ell(\phi, \nabla \phi) = G_c \left[\frac{(1 - \phi)^2}{2\ell} + \frac{\ell}{2} |\nabla \phi|^2 \right]. \quad (11)$$

Varying equation (11) produces the fracture terms in the adopted weak equation below. Thus this completion is traceable to the source weak form, not an unrelated crack-density model.

4.2 Normalized degradation

The source states $g(\phi) = \phi^2 + k_{\text{tol}}$, for which $g(1) = 1 + k_{\text{tol}}$. That slightly increases the stiffness of intact material. The implementation uses

$$g(\phi) = k_{\text{res}} + (1 - k_{\text{res}})\phi^2, \quad 0 < k_{\text{res}} \ll 1, \quad (12)$$

so that $g(0) = k_{\text{res}}$, $g(1) = 1$, and $g'(\phi) = 2(1 - k_{\text{res}})\phi$. The last identity explains the factor $(1 - k_{\text{res}})$ in the implemented phase driving term.

4.3 Adopted energy and spectral split

The completed instantaneous elastic-plus-fracture energy is

$$\mathcal{E}(\mathbf{u}, \phi) = \int_{\Omega} \left[g(\phi) \psi^+(\boldsymbol{\varepsilon}(\mathbf{u})) + \psi^-(\boldsymbol{\varepsilon}(\mathbf{u})) + G_c \left(\frac{(1 - \phi)^2}{2\ell} + \frac{\ell}{2} |\nabla \phi|^2 \right) \right] dV. \quad (13)$$

The source does not identify a split variant. The project supplies the 2D Miehe-type spectral strain split

$$\psi^\pm(\boldsymbol{\varepsilon}) = \frac{\lambda}{2} \langle \text{tr } \boldsymbol{\varepsilon} \rangle_\pm^2 + \mu \sum_{i=1}^2 \langle \varepsilon_i \rangle_\pm^2, \quad (14)$$

where ε_i are the principal strains. Machine-scale smoothing of the Macaulay brackets and eigenvalue discriminant makes the UFL residual differentiable. Plane strain is the default; plane stress is selectable through the effective 2D Lamé parameter.

4.4 Corrected phase weak form

For a fixed history field H , the implemented unconstrained scalar equation is: find $\phi \in V_\phi$ such that for every $q \in V_\phi$,

$$\int_{\Omega} \left[G_c \ell \nabla \phi \cdot \nabla q + \frac{G_c}{\ell} (\phi - 1) q + 2(1 - k_{\text{res}}) H \phi q \right] dV = 0. \quad (15)$$

Equivalently,

$$a_H(\phi, q) = \int_{\Omega} \left[G_c \ell \nabla \phi \cdot \nabla q + \left(\frac{G_c}{\ell} + 2(1 - k_{\text{res}}) H \right) \phi q \right] dV, \quad (16)$$

$$L(q) = \int_{\Omega} \frac{G_c}{\ell} q dV, \quad a_H(\phi, q) = L(q). \quad (17)$$

For $H \geq 0$, $G_c > 0$, and $\ell > 0$, the mass-reaction term and gradient term make the unconstrained operator symmetric positive definite after essential phase conditions are applied.

5 Acceptance and changes matrix

The matrix is deliberately more than a list of code differences. An **ACCEPTED** row confirms that the source concept remains part of the model; it does not imply that the source supplied every constitutive or numerical detail. A **CORRECTED** row identifies an incompatibility that would change the mathematical problem. A **COMPLETED** row supplies a missing definition or acceptance rule while preserving the source intent. A **SUPPLIED** row is a reproducibility or platform decision made solely by this project.

Severity and disposition are separate ideas. The sign conflict is critical because it changes the scalar operator. Missing Newton, KKT, convergence, and rollback rules are high-consequence completions because they determine whether a computed load state is admissible. Geometry and numerical defaults are project-supplied because the source never defines a boundary-value problem. This separation prevents benchmark choices from being presented as if they were facts extracted from the PDF.

Table 1: Traceable disposition of source statements and omissions.

ID	Source statement or omission	Disposition	Implemented decision	Technical rationale and evidence
A1	$\phi = 1$ intact and $\phi = 0$ broken.	ACCEPTED	Retain the intactness convention everywhere. Enforce $0 \leq \phi \leq 1$ and $\phi_{n+1} \leq \phi_n$.	This fixes the irreversibility direction and the AT2 local term. Bounds are verified at phase degrees of freedom.
A2	Split elastic energy; degrade only ψ^+ .	ACCEPTED	Use $g(\phi)\psi^+ + \psi^-$ and compute stress by differentiating this density with respect to strain.	Preserves compressive stiffness and the source's intended crack-face contact behavior at the constitutive level.
A3	Quasi-static momentum balance and weak mechanical residual.	ACCEPTED	Retain $\int_{\Omega} \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\delta \mathbf{u}) dV = 0$ for the displacement-controlled benchmark.	The post-phase free-DOF residual is reassembled and must satisfy the coupled acceptance tolerance.
A4	CG1 displacement, CG1 phase, DG0 history.	ACCEPTED	Use vector CG1, scalar CG1, and cellwise DG0 on affine triangles.	For CG1 affine triangles the strain and spectral energy are cellwise constant; DG0 is exact for the selected one-value-per-cell history representation.
A5	Staggered displacement–history–phase ordering.	ACCEPTED	Retain the ordering as the inner fixed-point architecture.	The ordering is appropriate, but a single pass is not treated as a converged load step.
C1	Strong Eq. (3) and weak Eq. (5) have incompatible local-term signs.	CORRECTED	Treat Eq. (5) as authoritative and use equation (15).	Integration by parts proves the mismatch. Eq. (5) yields the coercive positive diffusion–reaction operator; the printed strong form does not.
C2	$f_0(\phi)$ and c_w are not defined.	COMPLETED	Adopt $f_0 = (1 - \phi)^2$ and $c_w = 2$.	These AT2 choices are implied by Eq. (5); their variation reproduces the fracture portion of the implemented residual.
C3	$g(\phi) = \phi^2 + k_{\text{tol}}$.	CORRECTED	Use $k_{\text{res}} + (1 - k_{\text{res}})\phi^2$.	Restores exact intact stiffness, retains residual broken stiffness, and requires the factor $(1 - k_{\text{res}})$ in phase driving. Endpoint and monotonicity tests cover this choice.
C4	A spectral split is stated, but its variant and plane assumption are not.	COMPLETED	Use a smoothed 2D spectral strain split; plane strain by default and plane stress as an option.	Mixed-strain response is otherwise non-unique. Reference constitutive tests cover tension, compression, and the Lamé parameters.
C5	No nonlinear mechanics procedure is supplied.	COMPLETED	Differentiate the UFL residual for a consistent tangent and solve Newton corrections with residual, increment, and essential-BC checks.	The spectral split makes stress depend nonlinearly on displacement; a one-shot linear solve would be inconsistent. Failure is explicit and retryable.
C6	One staggered pass is shown per load increment.	CORRECTED	Repeat mechanics, transactional history, and constrained phase solves until phase increment, phase KKT residual, and post-phase free mechanical residual all pass.	This closes the coupling residual created when a newly damaged phase changes mechanical equilibrium. Nonconverged trials are never accepted.

Continued on next page

Table 1 continued from previous page

ID	Source statement or omission	Disposition	Implemented decision	Technical rationale and evidence
C7	History is said to enforce irreversibility, but no accepted-state rule is defined.	COMPLETED	Freeze (ϕ_n, H_n) while attempting a load. Rebuild $H = \max(H_n, \psi^+(u))$ on every staggered iteration and impose $0 \leq \phi \leq \phi_n$.	Prevents internal iterates or rejected loads from becoming fictitious physical history and makes nodal irreversibility explicit.
C8	No numerical method is supplied for the phase inequality.	COMPLETED	Solve the convex quadratic under box bounds. Use a clipped direct solution only as an initial candidate; call L-BFGS-B if necessary; require the projected KKT residual to pass.	Pointwise clipping alone is not an A -norm projection and generally does not satisfy constrained optimality. An automated test forces an active upper bound and optimizer path.
C9	The loop uses <code>while t <= Tmax</code> and increments t inside.	CORRECTED	Interpret <code>load_steps</code> as a nominal schedule, cap targets at pseudo-time 1, and store an explicit initial row at zero.	Avoids the off-by-one load beyond $T_{\max}$ and makes accepted pseudo-time strictly increasing with a terminal target of 1.
C10	No failed-step policy or recovery rule is stated.	COMPLETED	Snapshot fields and counters; on rejection restore all state, reduce the load increment, and retry. Recover a reduced increment only after several easy accepted steps.	Crack growth can cause abrupt loss of staggered convergence. Transactional rollback prevents contamination; terminal failure produces a partial auditable summary.
B1	No geometry, supports, loading, or initial flaw are supplied.	SUPPLIED	Use a unit-square single-edge-notched tension benchmark with bottom vertical support, one horizontal pin, top prescribed displacement, and an internal $\phi = 0$ starter notch.	The source is a model outline, not a reproducible boundary-value problem. These choices are explicitly project-supplied and mesh-aligned.
B2	No mesh/regularization relation is supplied.	SUPPLIED	Define $h = \sqrt{\Delta x^2 + \Delta y^2}$ as the largest triangular edge and, in strict mode, require $h/\ell \leq 0.5$.	Rejects visibly under-resolved diffuse cracks. Production claims still require a refinement and length-scale sensitivity study.
B3	The PDF requests only displacement and phase field output.	SUPPLIED	Add force, energy snapshots, damage and connected crack-front metrics, convergence/KKT data, cutback data, manifests, atomic summaries, plots, and XDMF/HDF5 fields.	A plausible contour alone is not numerical evidence. Only accepted states enter CSV and field files.
B4	No platform/backend is specified.	SUPPLIED	On native Windows, assemble with DOLFINx and solve serial sparse systems with SciPy; reject MPI runs larger than one rank.	This matches the installed backend, which lacks the PETSc path used by standard parallel DOLFINx workflows. It is an implementation constraint, not a model claim.

6 Nonlinear, constrained, accepted-state algorithm

6.1 Mechanics at fixed phase

Because equation (14) changes active principal-strain branches, the displacement residual is nonlinear. At Newton iteration m , the implementation solves

$$\mathbf{J}_u(\mathbf{u}^m; \phi) \Delta \mathbf{u}^m = -\mathbf{R}_u(\mathbf{u}^m; \phi), \quad \mathbf{u}^{m+1} = \mathbf{u}^m + \Delta \mathbf{u}^m, \quad (18)$$

with the Jacobian obtained by UFL differentiation. Convergence requires a small free-DOF residual, a small increment, and negligible prescribed-BC violation. After the phase update, the free mechanical residual is assembled again; this second residual is the relevant coupled acceptance check.

6.2 Transactional history update

Let $(\mathbf{u}_n, \phi_n, H_n)$ denote the last accepted load state. Within every staggered iteration j of the attempted next load, the trial history is rebuilt as

$$H^{(j)} = \max \left(H_n, \psi^+(\boldsymbol{\varepsilon}(\mathbf{u}^{(j)})) \right). \quad (19)$$

It is deliberately *not* computed as the maximum over previous internal iterations. Therefore an overshooting trial iterate does not become permanent physical history. Only when the load is accepted does $H^{(j)}$ become H_{n+1} .

6.3 Bound-constrained phase subproblem and KKT test

After finite-element assembly, the phase step has the convex quadratic form

$$\min_{\phi} J_H(\phi) = \frac{1}{2} \phi^T \mathbf{A}(H) \phi - \mathbf{b}^T \phi, \quad \mathbf{0} \leq \phi \leq \phi_n, \quad (20)$$

with notch degrees of freedom fixed to zero. Let $\mathbf{r} = \mathbf{A}\phi - \mathbf{b}$. The implemented projected residual has components

$$P_i(\mathbf{r}) = \begin{cases} 0, & l_i = u_i, \\ \min(r_i, 0), & \phi_i = l_i < u_i, \\ \max(r_i, 0), & \phi_i = u_i > l_i, \\ r_i, & l_i < \phi_i < u_i. \end{cases} \quad (21)$$

Thus a positive gradient at a lower bound and a negative gradient at an upper bound satisfy the KKT sign conditions and contribute zero. The load trial cannot be accepted unless $\|\mathbf{P}\|_\infty$ is below the configured phase KKT tolerance.

6.4 Coupled acceptance and adaptive load transaction

At a candidate target load, staggered iteration terminates successfully only if

$$\|\phi^{(j)} - \phi^{(j-1)}\|_\infty \leq \tau_\phi, \quad (22)$$

$$\|\mathbf{P}(\mathbf{A}\phi^{(j)} - \mathbf{b})\|_\infty \leq \tau_{\text{KKT}}, \quad (23)$$

$$\|\mathbf{R}_u(\mathbf{u}^{(j)}, \phi^{(j)})\|_{\infty, \text{free}} \leq \tau_{u, \text{coupled}}. \quad (24)$$

The implementation contract can be summarized as follows:

```

accepted state: (u_n, phi_n, H_n)
while accepted pseudo-time < 1:
    target = min(1, accepted time + current increment)
    snapshot u_n, phi_n, H_n and diagnostic counters
    apply target displacement
    repeat:
        Newton solve for u at fixed phi
        rebuild H = max(H_n, psi_plus(u))
        solve 0 <= phi <= phi_n with KKT check
        reassemble free mechanical residual using the new phi
    until all coupled tests pass or an iteration fails

    if converged:
        commit the trial; write one accepted record/frame
        consider increment growth after several easy accepted steps
    else:
        restore every snapshot; reduce the increment; retry

```

Rejected attempts do not create CSV rows or XDMF time frames. If consecutive cutbacks exceed the configured limit, the solver writes the last accepted state and a failed summary before raising an explicit error.

7 Benchmark choices supplied by the project

The source PDF contains no reproducible specimen. The following choices are therefore not claims about the source; they define the project's demonstration problem and defaults.

Item	Default	Role and provenance
Domain	1 mm × 1 mm	Unit-square 2D specimen; project-supplied.
Starter notch	0.20 mm from the left edge at mid-height	Internal nodal $\phi = 0$ constraint; diffuse crack, not a geometric slit.
Mesh	100 × 100 structured rectangles split into triangles	Even n_y and integer notch-length/ Δx are required for vertex alignment.
Elasticity	$E = 210 \text{ kN/mm}^2$, $\nu = 0.30$	Small-strain isotropic; plane strain default.
Fracture	$G_c = 2.7 \times 10^{-3} \text{ kN/mm}$, $\ell = 0.030 \text{ mm}$	AT2 parameters in the documented kN–mm system.
Residual stiffness	$k_{\text{res}} = 10^{-8}$	Prevents exact loss of ellipticity in fully damaged material.
Loading	Top u_y to 0.015 mm in 60 nominal increments	Bottom $u_y = 0$ plus one horizontal pin removes rigid motion.
Mesh safeguard	$h/\ell \leq 0.5$ with $h = \sqrt{\Delta x^2 + \Delta y^2}$	Pre-run resolution check; not a substitute for convergence studies.
Backend	Serial DOLFINx assembly plus SciPy sparse solves	Native-Windows installation constraint.

The starter notch is first equilibrated at zero load so that its diffuse AT2 profile is present before loading. The crack-front diagnostic follows only the notch-connected nodal set satisfying $\phi \leq 0.2$; it is mesh-quantized and is not used in the evolution law.

8 Verification and traceability evidence

8.1 Automated checks

The project test suite checks the following implementation contracts:

- normalized degradation endpoints and monotonicity;
- Lamé parameters and the spectral split in tension and compression;
- rejection of under-resolved meshes and misaligned starter notches;
- promotion of rank-deficient sparse solves to retryable failures;
- a phase QP that forces L-BFGS-B and an active irreversibility upper bound;
- an end-to-end FEniCSx run with finite diagnostics and readable XDMF;
- nondecreasing accepted history and damage measures;
- complete rollback after a rejected load trial; and
- auditable output after terminal nonlinear failure.

8.2 Recorded demonstration

The reduced, resolved demonstration under `results/demo/` is separate from the default production-sized configuration. Its manifest uses 30×30 cells per direction, $\ell = 0.1$ mm, $h/\ell = 0.4714$, and 45 nominal load increments. The recorded run completed all 45 accepted steps without a load cutback. Representative evidence from `results/demo/summary.json` is:

Quantity	Recorded value	Interpretation
Peak top reaction	0.6419596 kN at step 15	Demonstrates a peak and post-peak response; not an experimental validation target.
Final crack front / extension	1.0 mm / 0.8 mm	Connected threshold reaches the right boundary.
Final phase increment norm	8.78×10^{-6}	Below the configured staggered tolerance 10^{-5} .
Final projected KKT residual	8.03×10^{-10}	Below the configured absolute tolerance 10^{-8} .
Final free mechanical residual	9.98×10^{-8}	Below the configured coupled tolerance 10^{-6} .
Final phase optimizer activity	78 iterations; one active upper constraint	Confirms that the constrained optimizer and irreversibility path were exercised.

These results verify software behavior for one benchmark. They do not calibrate G_c or ℓ , establish mesh-independent fracture toughness, or validate the model against experimental data.

8.3 Source-to-code map

Source location	Implementation realization	Review disposition
PDF Eq. (1), free energy	<code>degradation</code> , spectral energy, elastic and fracture density forms	Completed with AT2 definitions and normalized degradation.
PDF Eq. (2), stress/equilibrium	Displacement residual, UFL-derived stress and Jacobian	Accepted; nonlinear Newton procedure supplied.
PDF Eq. (3), strong phase equation	Not implemented literally	Rejected where it conflicts with Eq. (5).
PDF Eq. (4), mechanical weak form	<code>form_u_residual</code> and <code>form_u_jacobian</code>	Accepted for displacement control.
PDF Eq. (5), phase weak form	<code>form_phi_bilinear</code> and <code>form_phi_linear</code>	Authoritative, normalized, and constrained.
Algorithm lines 2–6	Mesh, CG1/CG1/DG0 spaces, and field initialization	Accepted with precise DG0 interpretation.
Algorithm lines 15–21	Newton displacement, transactional history, phase QP	Accepted ordering; convergence and KKT contract added.
Algorithm line 23	XDMF displacement and phase output	Accepted and augmented with diagnostics, manifests, summaries, and plots.

9 Known limits and correct interpretation

1. **History-field approximation.** The history method is a common engineering approximation to the exact crack-irreversibility variational inequality. Within that approximation, the code additionally solves the nodal phase bounds as a KKT-checked constrained problem.
2. **Instantaneous energy is not the history potential.** The reported elastic-plus-fracture value is an instantaneous physical snapshot based on current strain. After local unloading it is not the stationarity functional of the history-driven phase subproblem and is not an exact energy-balance residual.
3. **Diffuse starter crack.** The internal $\phi = 0$ line is not a traction-free geometrical slit; the mesh remains topologically connected.
4. **Residual scaling.** The KKT and coupled mechanical tolerances are absolute assembled residuals calibrated for the documented kN–mm benchmark. They must be reconsidered after major changes of units, mesh density, or material scale.
5. **Serial backend.** The native-Windows implementation is serial and uses SciPy sparse solves. It is not the MPI-parallel PETSc workflow commonly used for large DOLFINx models.
6. **Crack-front metric.** The $\phi \leq 0.2$ connected threshold is a reporting diagnostic, depends on mesh resolution, and is not an exact crack length.
7. **Output-directory hygiene.** Optional files from an earlier run are not deleted when the same directory is reused with XDMF or plotting disabled. Fresh output directories should be used; the manifest and summary are the authoritative status/configuration records.
8. **Validation scope.** A successful demonstration and unit tests establish implementation behavior, not experimental validity. Quantitative conclusions require mesh and length-scale studies plus material calibration.

10 Reproduction and document build

The solver environment and commands are documented in `README.md`. This LaTeX document is standalone and can be rebuilt with TeX Live from the repository root:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \  
-outdir=output/pdf docs/PHASEFIELD_ALGORITHM_ACCEPTANCE_AND_CORRECTIONS.tex
```

Alternatively, change into `docs/` and run:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \  
-outdir=../output/pdf PHASEFIELD_ALGORITHM_ACCEPTANCE_AND_CORRECTIONS.tex
```

The PDF is expected at `output/pdf/PHASEFIELD_ALGORITHM_ACCEPTANCE_AND_CORRECTIONS.pdf`.

References

- [1] *Phase-Field Fracture: Governing Equations and FEniCS Implementation Algorithm*, two-page technical source supplied as `D:/Github/Codes/Crackpropagation/Karma/KarmaCrack.pdf`. No author or publication metadata is asserted because none is shown in the reviewed document.
- [2] *Critical review of the KarmaCrack phase-field algorithm*, project review record, `docs/ALGORITHM_REVIEW.md`.
- [3] *Corrected AT2 phase-field fracture benchmark implemented with FEniCSx*, project implementation, `phasefield_crack.py`, model version 1.0.0.
- [4] Project verification suite, `tests/test_phasefield_crack.py`.
- [5] Recorded demonstration manifest, accepted-state response, and summary under `results/demo/`.